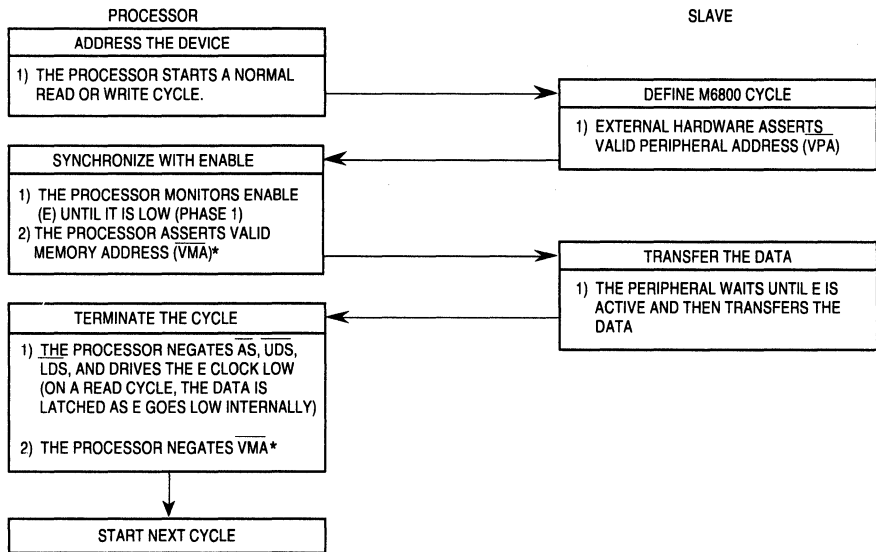


APPENDIX B M6800 PERIPHERAL INTERFACE

This appendix applies to all processors except the MC68EC000. Motorola's extensive line of M6800 peripherals is directly compatible with the M68000 Family. Since both the M6800 processors and the M68000 processors use memory-mapped I/O, interfacing the synchronous M6800 peripherals with the asynchronous M68000 Family processors works very well. The processor modifies its bus cycle to meet the M6800 cycle requirements whenever an M6800 device address is detected. Figure B-1 is a flowchart of the data transfer between the processor and M6800 devices.



* For MC68008-based systems, $\overline{VMA}$ is supplied by external circuitry.

Figure B-1. M6800 Data Transfer Flowchart

B.1 DATA TRANSFER OPERATION

Three signals on the processor provide the M6800 interface: enable (E), valid memory address ($\overline{VMA}$), and valid peripheral address ($\overline{VPA}$). Enable corresponds to the E or phase-two signal in M6800 systems. The bus frequency is one-tenth of the M68000 system clock frequency. The timing of E allows 1-MHz peripherals to be used with 8-MHz

processors. Enable has a 60/40 duty cycle; that is, it is low for six system clocks and high for four system clocks. This duty cycle allows $\overline{VPA}$ accesses on successive E pulses.

In the MC68000, MC68HC000, MC68HC001, and the MC68010, $\overline{VMA}$ is provided to indicate synchronization with E. The MC68008 does not provide a $\overline{VMA}$ signal; external circuitry similar to that shown in Figure B-2 using transistor-to-transistor (TTL) logic must be included in the system to provide $\overline{VMA}$. The $\overline{VMA}$ signal indicates to the M6800 devices that the address on the address bus is a valid device address and that the processor is synchronized to the enable clock. The VPA decode input is an active-high signal that is asserted when address strobe $\overline{AS}$ has been asserted and the address on the address bus is that of a peripheral device. The flip-flop on the left sets at the falling edge of E; the flip-flop on the right sets at the next fall of system clock, asserting $\overline{VMA}$. $\overline{VMA}$ remains asserted until the fall of system clock immediately following the negation of VPA decode. Figure B-3 shows the timing for the $\overline{VMA}$ signal provided by this circuitry.

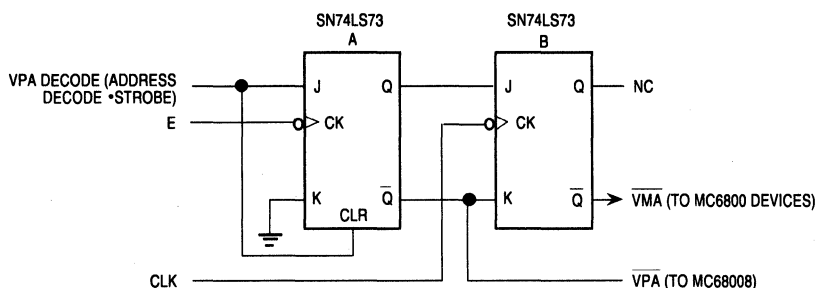


Figure B-2. Example External $\overline{VMA}$ Circuit

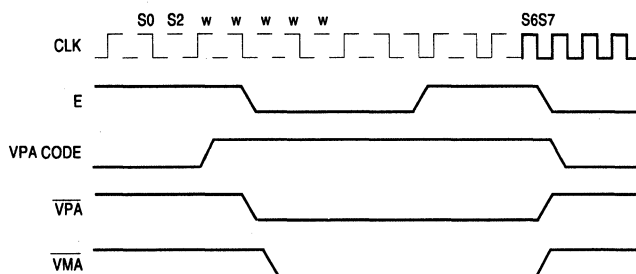


Figure B-3 External $\overline{VMA}$ Timing

M6800 cycle timing is shown in Figures B-4 and B-5. At state 0 (S0) in the cycle, the address bus is in the high-impedance state. A function code is asserted on the function code output lines. In state 1 (S1), the address is placed on the address bus. During state 2 (S2), the address strobe ($\overline{AS}$) is asserted to indicate that the address on the bus is valid. If the bus cycle is a read cycle, the upper and/or lower data strobe ($\overline{UDS}$, $\overline{LDS}$) (MC68000/MC68HC000/MC68HC001/MC68010) or data strobe ($\overline{DS}$) (MC68008) is also

asserted in S2. If the bus cycle is a write cycle, the read/write ($R/\overline{W}$) signal is driven low (for write) during S2. In state 3 (S3) of a write cycle, the write data is placed on the data bus, and in state 4 (S4), the data strobes are asserted to indicate valid data on the bus. Next, the processor inserts wait states until it recognizes the assertion of $\overline{VPA}$.

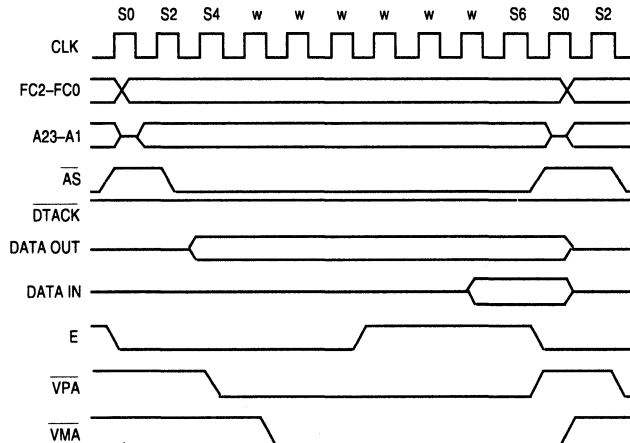


Figure B-4 M6800 Peripheral Timing—Best Case

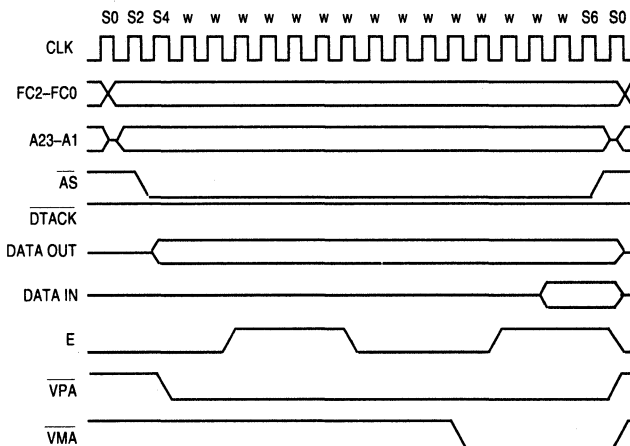


Figure B-5. M6800 Peripheral Timing—Worst Case

The $\overline{VPA}$ input indicates to the processor that the address on the bus is the address of an M6800 device (or an area reserved for M6800 devices) and that the bus should conform to the phase-two transfer characteristics of the M6800 bus. $\overline{VPA}$ is derived by decoding the address bus, conditioned by the address strobe ($\overline{AS}$).

B

After recognizing $\overline{VPA}$, the processor assures that enable (E) is low by waiting, if necessary, and subsequently asserts $\overline{VMA}$. $\overline{VMA}$ is then used as part of the chip-select equation of the peripheral to ensure correct timing for selection and deselection of the M6800 device. Once selected, the peripheral runs its cycle during the high portion of the E signal. Figure B-4 shows the best-case timing of an M6800 cycle, and Figure B-5 shows the worst-case timing. The cycle length is entirely dependent on the relationship of the assertion of $\overline{VPA}$ to the E clock.

When external circuitry asserts $\overline{VPA}$ as soon as possible following the assertion of $\overline{AS}$, the assertion of $\overline{VPA}$ is recognized on the falling edge of S4. In this case, no extra wait states are inserted (waiting for the assertion of $\overline{VPA}$). The only wait states inserted are those required to synchronize with the E clock. The synchronization delay is an integral number of system clock cycles within the following extremes:

1. Best Case—the assertion of $\overline{VPA}$ is recognized on the falling edge three clock cycles before E rises (or three clock cycles after E falls).
2. Worst Case—the assertion of $\overline{VPA}$ is recognized on the falling edge two clock cycles before E rises (or four clock cycles after E falls).

The processor latches the peripheral data in state 6 (S6) during a read cycle. For all cycles, the processor negates the address and data strobes one-half clock cycle later in state 7 (S7), and E goes low at this time. Another half clock later, the address bus is placed in the high-impedance state, and $R/\overline{W}$ is driven high. Logic in the peripheral must remove $\overline{VPA}$ within one clock after the negation of address strobe.

Data transfer acknowledge ($\overline{DTACK}$) must not be asserted while $\overline{VPA}$ is asserted. The state machine in the processor looks for $\overline{DTACK}$ to identify an asynchronous bus cycle and for $\overline{VPA}$ to identify a synchronous peripheral bus cycle. If both signals are asserted, the operation of the state machine is unpredictable.

To allow the processor to place its buses in the high-impedance state during DMA requests without inadvertently selecting the peripherals, $\overline{VMA}$ is active low for the M68000 Family of processors. The active-low $\overline{VMA}$ is in contrast to the active-high $\overline{VMA}$ signal of the M6800.

B.2 INTERRUPT INTERFACE OPERATION

During an interrupt acknowledge cycle while the processor is fetching the vector, $\overline{VPA}$ is asserted, and the processor (or external circuitry) asserts $\overline{VMA}$ and completes a normal M6800 read cycle as shown in Figure B-6. For the interrupt vector, the processor uses an internally generated vector number called an autovector. The autovector corresponds to the interrupt level being serviced. The seven autovectors are decimal vector numbers 25–31.

The autovector operation, which can be used with all peripherals, is similar to the normal interrupt acknowledge cycle. The autovector capability provides vectors for each of the six maskable interrupt levels and for the nonmaskable interrupt level. Whether the device supplies the vector number or the processor generates an autovector number, the

interrupt service routine can be located anywhere within the supervisor program address space because the user assigns the vectors in the vector table.

Since $\overline{VMA}$ is asserted during an autovector operation, care should be taken to prevent an unintended access to the device. An unintended access could occur if the peripheral address were on the address bus during the autovector operation.

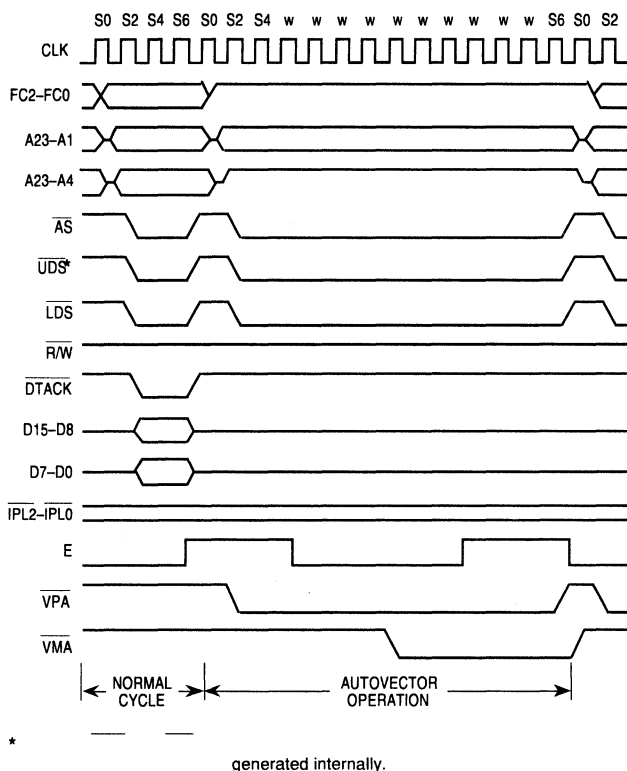


Figure B-6. Autovector Operation Timing Diagram

